Subhrajit **Prusty**

LEAD BACKEND ENGINEER

Bengaluru, India

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Summary

Backend & infrastructure engineer with 5+ years shipping and operating high-availability systems in Go and Python on AWS and Kubernetes (EKS). Build distributed services from 0 to 1 and operate them at scale — tens of millions of calls a month and ~99.5% uptime — backed by CI/CD, autoscaling, and observability that keep them reliable and cost-efficient. Deep hands-on experience with real-time voice AI and LLM-powered systems.

Experience

Bolna

Bengaluru, India

LEAD BACKEND ENGINEER

January, 2026 - Present

- Own the platform infrastructure behind a voice-AI product handling ~34M calls a month across ~2,400 tenants in Ecommerce, EdTech, HealthTech, BFSI, and Hospitality — multi-region Kubernetes (API, WebSocket, and GPU-inference clusters) scaled ~6x in six months; own capacity, autoscaling (Cluster Autoscaler, Karpenter, and custom-metric/queue-depth HPAs on real application signals via DatadogMetric), and ALB routing tuned for latency-sensitive WebSocket and SIP traffic, plus the cloud bill — a five-figure monthly AWS spend cut ~10% in a single pass with ~70% less node churn
- Own the security posture — a least-privilege, federated-identity model across every workload, host, and CI pipeline (per-workload IRSA, per-instance EC2 instance profiles, OIDC-federated CI), with centralized secrets via External Secrets Operator, policy-simulation-verified RBAC, and a hardened production boundary; ISO-aligned
- Own the self-hosted ASR/GPU-inference platform that every live call depends on — GPU inference running on ~60% of all cluster compute, owned end to end: model lifecycle, autoscaling to real-time stream capacity, licensing and model storage, and a multi-cloud failover path built to survive single-cloud GPU stockouts
- Own the on-call and incident-management process for a platform where downtime is lost revenue — run incident response end to end on the monitoring and alerting foundation I own (Datadog metrics and APM, self-hosted Loki + Grafana, OpenTelemetry tracing), commanding multi-hour cross-stack incidents and building the diagnostic fingerprints that separate a platform-side issue from a telephony/ASR vendor outage
- Work an infra-down loop into the backend — trace infra-level symptoms (OOM-killed pods, stalled event loops, connection-pool exhaustion, cost regressions) to their root cause in application code and fix both layers: cut a queue worker's per-pod memory ~95% (from ~15GB to under 1GB), hardened the WebSocket fleet's health signal against event-loop stalls, and cut LLM client p99 latency from ~231s to ~55s under +54% traffic
- Re-architected post-call processing (CDR, settlement, disposition analytics) into an idempotent, self-reconciling pipeline — atomic claim-and-checkpoint, reconciliation jobs that recover interrupted calls, and bounded provider retries — giving exactly-once settlement and safe horizontal scale across concurrent, multi-pod execution
- Built kubedump, an open-source Go CLI that snapshots live cluster state into clean, Git-trackable manifests on a daily CI schedule — the backbone for config drift-tracking and audit trails across all clusters

skit.ai

Bengaluru, India

SOFTWARE ENGINEER - PRODUCT

November 2022 - August 2025

- Engineered an LLM-powered voicebot in Go, processing millions of calls for 40+ enterprise clients.
- Rebuilt Voicebot on the open-source Pipecat-AI framework, enabling modularity and third-party integration.
- Designed and implemented a bidirectional audio-streaming telephony stack with gRPC for real-time interactions.
- Configured and maintained in-house + OpenSIPS infrastructure for call routing with telephony vendors like TCN and Twilio.
- Designed and implemented a flow-configuration validation service, completely eliminating config-related incidents.
- Maintained and enhanced the voicebot dialogue management system, adding features and improving reliability.
- Optimized flow-configuration loading, reducing memory usage by 80%.
- Served as primary on-call engineer managing incident response for critical voicebot and telephony infrastructure.
- Maintained 99.5% uptime for voicebot systems using Grafana, Loki, OpenTelemetry, Prometheus for monitoring.

skit.ai *Bengaluru, India*

SOFTWARE ENGINEER - SOLUTIONS *December 2020 - October 2022*

- Owned and maintained a critical internal repository, establishing code review processes and authoring contribution guidelines; significantly improving code quality and reducing technical debt.
- Designed and optimized voice bot flow configurations in collaboration with cross-functional teams.
- Developed 3rd-party API integrations, including PCI-DSS compliant Payment APIs, improving voice bot's debt collection capabilities
- Enhanced build pipelines, reducing deployment time and improving CI/CD efficiency.
- Maintained, customized and enhanced internal ML models used for real time conversational voice bots.
- Designed and implemented an automated daily reports system that simplified accessing and transforming on-prem data

racetrack.ai (Internship) *Bengaluru, India*

RESEARCH AND DEVELOPMENT TEAM *August 2019 - March 2020*

- Using data collected by automated web-scraping, published an abstract on Personalized Hybrid-Recommender Systems using social network analysis.
- Using realtime image processing, developed a minor claims automation service for car accident processing, improving claim resolution efficiency.
- Maintained and improved the performance of the primary AI chatbot product, Marvin.

ITC Infotech, Innoruption Lab (Internship) *Bengaluru, India*

RESEARCH AND DEVELOPMENT TEAM *May 2019 - July 2019*

- Using image processing and automation, developed a metal sheet counting algorithm.
- Using facial detection and recognition, built an ID-card detail extraction system, optimizing concierge check-in process.

Education

Institute of Technical Education and Research (ITER) *Bhubaneswar, Odisha, India*

B.TECH IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY *2016 - 2020*

- CGPA : 8.81

Ispat English Medium School (IEMS) *Rourkela, Odisha, India*

INTERMEDIATE/+2 FROM ISC *2014 - 2016*

- Percentage : 84.5

Ispat English Medium School *Rourkela, Odisha, India*

MATRICULATION FROM ICSE *2014*

- Percentage : 94.17

Skills

Languages	Python, Go, Bash, SQL
Infra & Cloud	Kubernetes, Docker, Helm, External Secrets Operator, GPU instances, AWS (EC2, IAM/IRSA, OIDC, Secrets Manager, S3, ALB, Lambda, EFS), GitHub Actions
Observability	Datadog, Grafana, Loki, Prometheus, OpenTelemetry, ClickHouse, ClickStack
Backend & Messaging	FastAPI, Flask, gRPC, WebSockets, RabbitMQ, Redis, PostgreSQL, SIP/OpenSIPS